SRIPAD SIRIKONDA — Software Engineer / ML Engineer - SF Compute

US Citizen · Open to all Locations · sripadsirik@gmail.com · linkedin.com/in/sripapa-cs/ · github.com/sripadsirik · sripapa.vercel.app

Education

University of Illinois at Chicago *Expected May 2026*
B.S. in Computer Science GPA: 3.8/4.0
AWS Solutions Architect Associate (In Progress)

Skills

Languages: Python, C++, C, Java, Swift, Rust, JavaScript, SQL
Frameworks: React, Next.js, Spring Boot, FastAPI, Flask, Node.js
Cloud & DevOps: AWS, GCP, Docker, Kubernetes, CI/CD, Linux
Data & ML: PostgreSQL, Redis, gRPC, TensorFlow, LangChain, LLM APIs (OpenAI, Groq)
Systems: Distributed Systems, Microservices, Backend APIs, Data Modeling, Control Planes, State Reconciliation, Idempotent Systems

Experience

Adaptive Robotics Lab, UIC — Embedded Systems Engineer Intern Chicago, IL *2025 – Present*

- Built vision-based perception modules in Python and C++ to estimate contact forces and interaction points, improving manipulation robustness in experimental trials.
- Developed kinematics and simulation pipelines using URDF, forward/inverse kinematics, and physics tooling to validate a 7-DOF manipulator across 100+ configurations, reducing on-hardware testing by 25%.
- Designed modular robotics software for sensor processing and experiment reproducibility, implementing performance-critical components in C++ and Rust, supporting 50+ experimental runs and cutting setup/debug time by 30%.

Family Eye Physicians — AI Engineer Intern Chicago, IL *2025*

- Built and integrated AI-driven patient workflow systems on iOS, incorporating on-device inference and location-aware feature generation to improve data quality by 35% and reduce end-to-end ingestion latency by 30%.
- Designed production-grade data pipelines ingesting 100K+ daily sensor and interaction events via Pub/Sub–backed ETL and REST services, generating structured, model-ready features for downstream analytics and learning workflows.
- Deployed and monitored ML-enabled services using CI/CD pipelines with canary releases, automated validation, and observability metrics, reducing production incidents by 96% and improving model rollout stability.

SWIM Research Labs, UIC — Machine Learning Engineer Intern Chicago, IL *2024 – Present*

- Built parallelized pipelines processing 10K+ sensor readings daily, enabling near-real-time anomaly detection and inference at scale.
- Developed probabilistic inference and sensor fusion models, improving detection accuracy by 25% and reducing latency by 40%.
- Applied Bayesian optimization to evaluate and optimize sensor placement strategies, benchmarking performance against genetic algorithms to improve coverage efficiency and resource utilization by 35%.

Chicago Saranam Yatra Organization — Software Engineer (Mobile) Aurora, IL *2023 – 2024*

- Shipped a cross-platform booking app using React, Swift (iOS), Spring Boot, Firebase, and GCP, serving 1K+ users with 99.9% uptime.
- Built GPS event check-ins with Core Location and native Swift modules, backed by Spring Boot services, reducing attendance errors by 50%.
- Optimized client-side data access with caching and state management, coordinating Spring Boot endpoints to improve responsiveness by 40%.

Projects

AI-Driven Asset, Contract & Allocation Platform — Founding Startup *Dec 2025*

- Architected and implemented an AI-driven real estate intelligence system with normalized property and portfolio schemas, deterministic cash-flow simulations, and auditable decision logic, backed by a FastAPI and gRPC services integrating Groq LLMs via LangChain to perform context-aware reasoning over live asset data at enterprise scale.

Tesla Battery Optimizer *Mar 2025*

- Built an end-to-end machine learning system predicting battery State of Charge (SOC) with 95% accuracy using sensor telemetry and feature engineering, reducing charge time by 15%, while developing a real-time SwiftUI dashboard for 500+ battery modules and optimizing inference via batching and AWS Lambda autoscaling to reduce compute costs by 20%.

Distributed Market & Allocation Systems Platform - SparkHacks Winner *Feb 2025*

- Designed and scaled a fault-tolerant paper trading platform using queue-driven ingestion and idempotent, state-reconciled transaction handling, processing 5K+ simulated trades per month with 99.9% uptime, while deploying A/B-tested trading flows that increased user retention by 25% and supporting 100+ concurrent users with retry-safe, zero-downtime event handling.

NASA Planetary Seismic Detector *Nov 2024*

- Developed a machine learning pipeline processing 1,200+ hours of planetary seismic data to detect 850+ events, improving accuracy by 30% through signal preprocessing and supervised tuning, and reducing false positives by 20% via Bayesian data fusion.